

FLYBASE REAGENT STOCKER

From Gene Sets to Fly Stocks

End-to-end prioritization of Drosophila reagents for sleep/circadian follow-up

22

input gene-set tables

3

source families

6

priority buckets

8,300+

stocks organized

What this stocker run produces

22

input gene-set tables

619

targeted input genes

5796

transcriptomics stocks

6

priority buckets per set

Core message

The pipeline turns heterogeneous biological evidence into orderable, phenotype-backed fly reagents, preserving source provenance and literature support so follow-up choices are auditable.

Narrative flow

The deck starts with source provenance and prioritization logic, then moves into examples and deeper dataset-specific evidence.



Targeted runs

GABA/vGAT screen, bipolar GWAS orthologs, AD orthologs, and curated Slumber/Jordan lists.



Transcriptomics run

Rosensweig/Shah sleep-wake manuscript-derived gene sets: CSW tiers, sleep-history/rebound correlations, and homeostatic candidates.



Output

Every input table becomes a workbook with six priority sheets, references, all phenotypic stocks, and count summaries.

What happens to each input gene set

1

Input gene sets

22 tables: screen hits, human GWAS orthologs, curated candidates, transcriptomics tiers

2

FlyBase stock expansion

For each FBgn, collect stocks, alleles, constructs, genotypes, phenotypes, and references

3

Phenotype + literature gates

Keep phenotype-backed reagents; flag sleep/circadian/rhythm/locomotor support

4

Priority sorting

Assign each stock once into six orderability/reagent buckets

5

Review-ready workbook

Six bucket sheets + references + all phenotypic stocks + contents/counts

Priority buckets: orderability first, then reagent class; each stock appears once

Bloomington • UAS / RNAi

Stock-center, overexpression / knockdown

Vienna (VDRC) • UAS / RNAi

VDRC knockdown collection

Bloomington • Allele / Insertion

Stock-center classical alleles

Stock Center • Allele / Insertion

Other orderable centers (Kyoto, etc.)

Non-Stock-Center • UAS / RNAi

Custom phenotype reagents (no FBst)

Non-Stock-Center • Allele / Insertion

Custom phenotype reagents (no FBst)

The practical question this answers: for any candidate gene, which phenotype-backed fly reagents are easiest to order and most relevant to sleep/circadian biology?

Three source families feed the same prioritization workflow

Screen-derived

GABA gene list from Aldeb bulk-seq vGAT hits (bottom 10%).
Direct fly genes with FBgn IDs.

Human genetics

Bipolar and Alzheimer's disease human gene sets mapped to Drosophila through DIOPT, retaining orthology confidence.

Lab/curated sleep biology

Jordan VCM, Slumber neuropeptide/transmitter, Slumber RNAi, and Rosensweig/Shah transcriptomics-derived tiers.

Why the source split matters

- **Direct fly gene sets**
No orthology step; stocker starts from FBgn IDs and expands to available reagents.
- **Human disease gene sets**
Interpretation depends on orthology confidence and source study definition; DIOPT score/rank are therefore shown.
- **Transcriptomics tiers**
Genes are evidence-ranked by recurrence across sleep-wake manipulations or by history/rebound correlations.

All three source families converge to the same stock workbook format, so downstream comparison is apples-to-apples.

Targeted gene-list inputs

Dataset	Reference	Dataset description	Stocker input
vGAT screen	Aldeb bulk-seq vGAT hits	Internal screen-derived gene list; bottom-10% vGAT hits, supplied as symbols + FBgn IDs.	140 fly genes directly stocked.
Bipolar GWAS	O'Connell, Koromina, van der Veen et al.; published 22 Jan 2025; Nature 639:968-975.	Human bipolar-disorder GWAS: 158,036 BD cases + 2.8M controls; 298 loci. Supplementary Table S31 was mapped through DIOPT and Filter2.	168 fly ortholog rows; DIOPT score/rank retained.
Alzheimer's GWAS	Bellenguez/Lambert AD GWAS-derived ortholog list.	Human Alzheimer's disease candidate genes mapped to Drosophila orthologs and paired with pre-pulled BDSC RNAi hints.	227 fly genes with DIOPT confidence + stock hints.
Jordan VCM	Internal curated candidate list.	Focused VCM candidate genes supplied as current symbols + FBgn IDs.	35 fly genes directly stocked.
Slumber NP/transmitters	Internal curated Slumber list.	Neuropeptide and transmitter genes likely relevant to sleep/wake phenotypes.	33 fly genes directly stocked.
Slumber RNAi	Internal curated Slumber list.	Short RNAi candidate list for targeted follow-up.	16 fly genes directly stocked.

Bipolar GWAS is now separated into reference metadata plus the downstream DIOPT-derived fly-gene table used by this run.

Rosensweig/Shah transcriptomics inputs

Shared Transcriptomic Responses to Distinct Sleep-Wake Manipulations Reveals Multiple Homeostatic Pathways in Drosophila

Rosensweig, Shah, Sisobhan, Andreani & Allada; posted 6 Jun 2026; bioRxiv preprint / lab manuscript submitted for review; doi: 10.64898/2026.02.28.708752

How the manuscript datasets map into stocker inputs

Baseline time course

Whole-brain RNA-seq at ZT0/4/8/12/16/20; validates clock transcript cycling.

Mechanical SD

3h, 6h, and 12h sleep deprivation ending at ZT0; each followed by rebound assessment.

Thermogenetic sleep

R85C10-GAL4 > TrpA1 activation promoted sleep and subsequent anti-rebound.

Published induction sets

Optogenetic R23E1040 > ChRimson and pharmacologic THIP whole-brain RNA-seq.

Correlation extension

Sleep-history/rebound correlations across 9 datasets; added WakeGAL4 TrpA1 lines and excluded opto/THIP where

Staged evidence

CSW FC1 and FC0.5 tiers ask: which genes respond consistently across independent sleep/wake manipulations? Correlation tiers ask: which genes track prior sleep history or predict rebound? fl.ai columns retain literature support.

Interpretation note

Transcriptomics gene counts can recur across tiers; stock counts are the cleaner headline metric.

How many stocks were organized

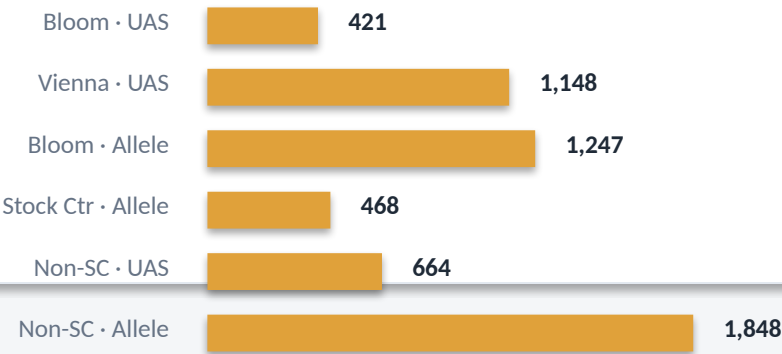
Targeted gene-list runs

Dataset	Genes	Stocks	Alleles	w/ stocks
vGAT — GABA gene list	140	2048	1497	135
BPD-GWAS orthologs	168	1633	1485	155
DIOPT Bellenguez/Lambert (AD)	227	2620	1767	217
Jordan VCM	35	757	692	34
Slumber neuropeptides	33	1156	523	32
Slumber RNAi candidates	16	225	173	16
All targeted	619	8439	6137	-

Transcriptomics run

Dataset category	Tiers	Stocks
CSW FC0.5	7	4025
CSW FC1	4	356
Homeostatic genes	1	46
History x Rebound overlaps	2	108
SleepHistory correlates	2	1261

Priority bucket load (all transcriptomics tiers)



Counts are de-duplicated within each workbook; transcriptomics genes may repeat across evidence tiers.

How a human GWAS gene becomes a fly stock shortlist

1

Human gene

MED24 from Nature 2025 bipolar GWAS

2

Orthology

DIOPT maps MED24 to fly MED24 (FBgn0035851), high confidence

3

Stock expansion

Find FlyBase stocks/alleles/constructs tied to MED24

4

Priority placement

Place orderable phenotype-backed reagents before custom/non-stock-center reagents

Input row

MED24 appears in the bipolar GWAS-derived DIOPT table. The source row retains human symbol, fly ortholog symbol, FBgn, DIOPT score, weighted score, rank, and support methods.

Interpretation

The detailed evidence and provenance remain traceable while the practical output is a short, ordered reagent list.

Representative GABA bucket examples still appear next, showing what the final stock-level rows look like.

One example reagent from each category

Representative rows from the GABA gene-list workbook (n=140) — every dataset follows the same six-table structure.

Bloomington • UAS • Phenotype

220 stocks in this bucket

ID: FBst0600261

Collection: Bloomington

Gene: Act5C; Gat

Genotype: $\gamma[1] w[1118]; P\{w[+mC]=Act5C-GAL4\}25FO1, P\{w[+mC]=UAS-GFP.U\}2/...$

Vienna (VDRC) • UAS • Phenotype

182 stocks in this bucket

ID: FBst0479639

Collection: Vienna

Gene: ATPsynCF6

Genotype: $P\{KK108581\}VIE-260B$

Bloomington • Allele/Insertion • Phenotype

568 stocks in this bucket

ID: FBst0003954

Collection: Bloomington

Gene: Act5C

Genotype: $\gamma[1] w[*]; P\{w[+mC]=Act5C-GAL4\}17bFO1/TM6B, Tb[1]$

Stock Center • Allele/Insertion • Phenotype

173 stocks in this bucket

ID: FBst0306375

Collection: Kyoto

Gene: Act5C

Genotype: $\gamma[1] w[*]; P\{w[+mC]=Act5C-GAL4\}25FO1/CyO, \gamma[+]$

Non-Stock-Center • UAS • Phenotype

225 stocks in this bucket

ID: Scer\GAL4[Act5C.PU]

Collection: Custom phenotype reagent

Gene: Act5C

Genotype: $Hsap\SOD1[UAS.cPa] Scer\GAL4[Act5C.PU]; Mmus\Prkaca[mC.UAS] Sc...$

Non-Stock-Center • Allele/Insertion • Phenotype

680 stocks in this bucket

ID: Cep290[mecH]

Collection: Custom phenotype reagent

Gene: Cep290

Genotype: Cep290[mecH]

Anatomy of every generated workbook

What each workbook contains

Sheet 1 – 6

One per priority bucket (color-coded above)

References

Supporting publications per stock / allele

All Phenotypic Stocks

Full union of phenotype-backed stocks

Contents

Prioritization rules + per-sheet count breakdown

Priority rule: a stock is placed in the first bucket it qualifies for, so the six tables never double-count a stock.

Key columns (approx. 59 total), grouped



Identity

FBst, stock_number, collection, genotype



Reagent type

RNAi, UAS, GAL4, mutant, sgRNA, custom_stock



Gene / allele

gene_symbols, flybase_gene_ids, AlleleSymbol, class



Construct parts

FBti / FBtp / FBal symbols + IDs, Balancers



Literature

PMID, total_refs, sleep/circadian ref counts



Phenotype + similarity

PHENOTYPE_RELEVANCE_SCORE, cosine similarities

How targeted inputs differ before stock expansion

Input family	Biological unit	Important fields retained	Why it matters
GABA/vGAT	Fly gene list	Gene, primary_symbol, FBgn, source dataset	Direct stock expansion
Bipolar GWAS	Human genes -> DIOPT	Human symbol, fly ortholog, FBgn, DIOPT score/rank, support tools	Preserve orthology confidence
AD GWAS	Human genes -> DIOPT	Human symbol, fly ortholog, FBgn, DIOPT rank, BDSC RNAi hints	Use disease orthologs as candidate fly genes
Jordan/Slumber	Curated fly lists	Submitted symbol, current symbol, FBgn, corrections	Focused candidate follow-up

Why provenance comes first

The first question is source and confidence. Column-level schema is useful only after the biological meaning and provenance are clear.

What remains auditable

The generated workbooks still retain the detailed source columns and stock-level FlyBase fields needed to trace a reagent back to the input evidence.

Transcriptomics: consistent sleep/wake response tiers

CSW FC1

Higher-confidence, stricter log2 fold-change threshold (>1). Captures genes repeatedly changing with sleep/wake state across datasets.

CSW FC0.5

Broader recurrence tier using a lower log2 fold-change threshold (>0.5) while preserving statistical rigor; useful for distributed homeostatic pathways.

CSW tier	Genes
FC1 Sleep freq 2	16
FC1 Wake freq 2	77
FC1 Wake freq 3	14
FC1 Wake freq 4	7
FC0.5 Sleep freq 2	135
FC0.5 Sleep freq 3	6
FC0.5 Wake freq 2	353
FC0.5 Wake freq 3	197

CSW tier	Genes
FC0.5 Wake freq 4	82
FC0.5 Wake freq 5	14
FC0.5 Wake freq 6	1

Frequency = number of datasets showing consistent sleep/wake direction; higher frequency means more reproducible transcriptomic support.

Transcriptomics: history, rebound, and literature support

Evidence layer	Size	Interpretation
SleepHistory correlates	215 genes	Genes whose expression correlates with prior sleep/wake history across time bins.
History x Rebound overlaps	20 genes	Genes linking prior history direction with predicted rebound direction.
Homeostatic genes	6 genes	Small focused set called out as sleep-homeostasis candidates.
fl.ai annotations	per gene	AI-assisted literature category, confidence, rationale, supporting references, and source papers.

Correlation logic

The paper asks whether expression tracks sleep-wake history before sampling and/or predicts rebound after sampling, using Pearson correlations against log2-normalized gene expression.

How this affects stock choice

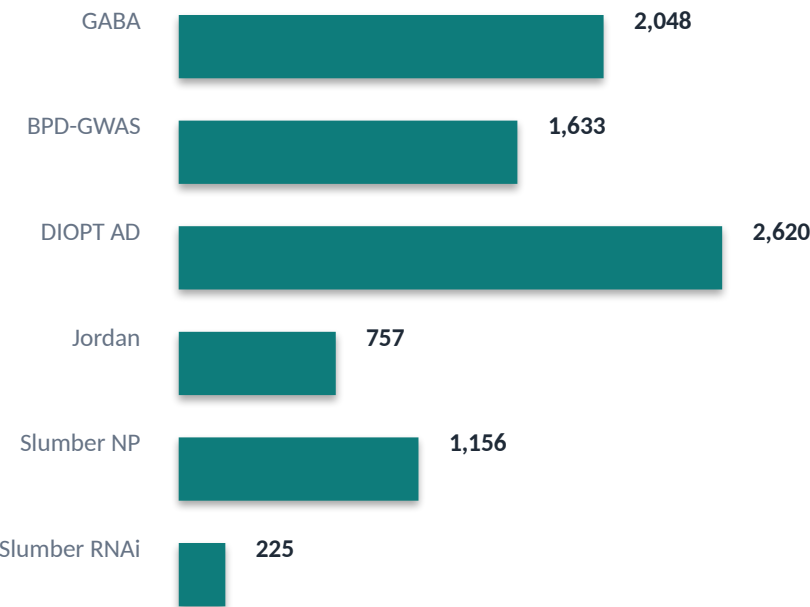
A gene that is both transcriptomically recurrent and literature-supported is easier to justify for reagent ordering than a single-method hit with no sleep/circadian evidence.

A fast decision path

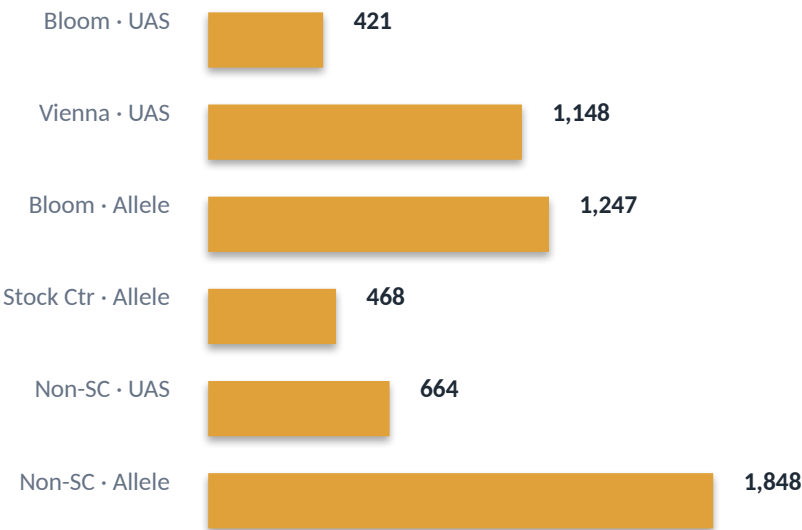
- **Start with source confidence**
Was this gene from direct fly evidence, a human GWAS ortholog, or a transcriptomics recurrence/correlation tier?
- **Check orderability**
Prefer Bloomington/Vienna stock-center reagents before non-stock-center or custom reagents.
- **Check reagent type**
Separate UAS/RNAi perturbation from allele/insertion reagents depending on the experiment.
- **Check biological support**
Use reference counts, sleep/circadian flags, phenotype relevance, and fl.ai rationale to prioritize follow-up.
- **Shortlist manually**
Open the relevant workbook sheet and review genotype/construct details before ordering.

Where the strongest reagent pools are

Targeted runs: stocks per dataset



Transcriptomics: stocks per bucket



This is the slide to use for a quick sense of scale before opening individual workbooks.

TAKEAWAYS

What the tables give you

1

22 gene sets, one consistent format

Every input list becomes the same six prioritized, phenotype-filtered stock tables.

2

Ordering by orderability + reagent type

Bloomington / Vienna / other centers first; UAS vs classical alleles separated; custom reagents last.

3

Built-in literature relevance

Sleep / circadian / rhythm / locomotor references and phenotype-similarity scores travel with each stock.

4

8,300+ phenotype-backed stocks organized

Ready to filter down to an orderable shortlist per gene.